UXR Receiver Analysis, Example from Mock Captures

MOCK/SIMULATED CAPTURES: NOT LABORATORY MEASUREMENTS

Data source of the captures: simulated. The label is the one each sidecar declares; this report cannot verify it.

4 test captures were analysed. Training captures (0) are used for fitting only and are never scored. Software version 1.2.0. Generated by opticps uxr-report from the ingest-uxr and compare-uxr outputs.

The simulated captures come from a mock receiver: a simulated link and no instrument. Their numbers show the report format and say nothing about hardware.

Simulation for comparison: results/synthetic/demo_results.json (data source: simulated, never a measurement).

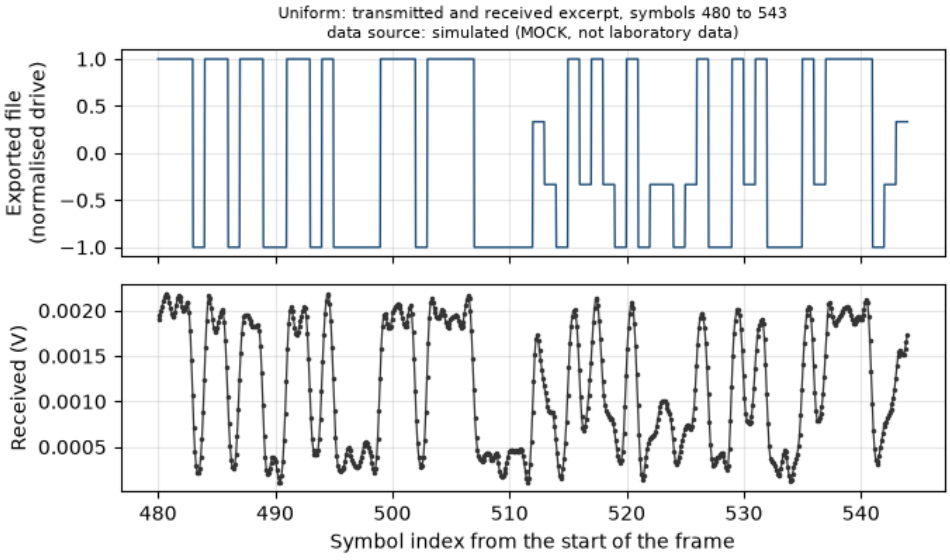
Uniform (PAM4_Uniform)

Error rates

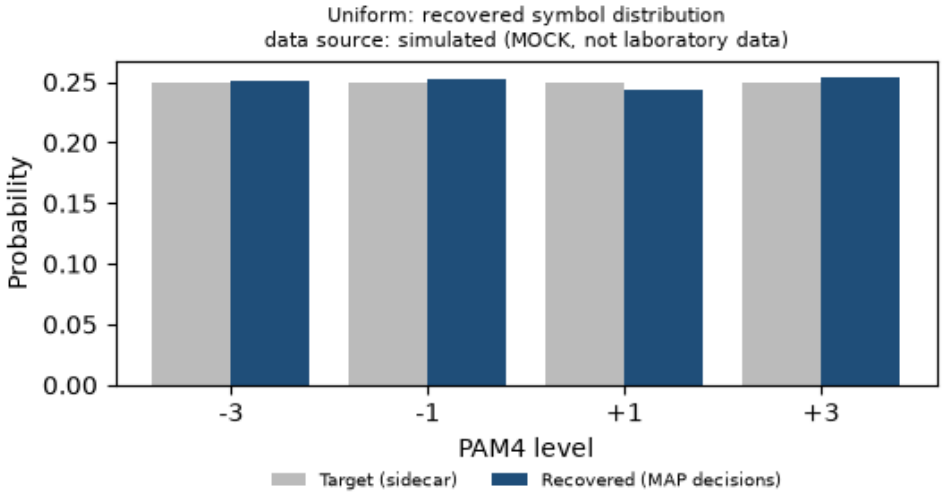
Capture	Condition	SER	BER
mock-PAM4_Uniform-r1	attenuation 18.4 dB, fiber 0 km, run r1	12 errors in 126976 symbols, rate 9.45e-05 (95% interval 4.88e-05 to 0.000165); 4 frames repeat the same 31744 payload symbols	12 errors in 253952 bits, rate 4.73e-05 (95% interval 2.44e-05 to 8.25e-05); 4 frames repeat the same 63488 payload bits

Post-matcher BER is diagnostic only; there is no FEC.

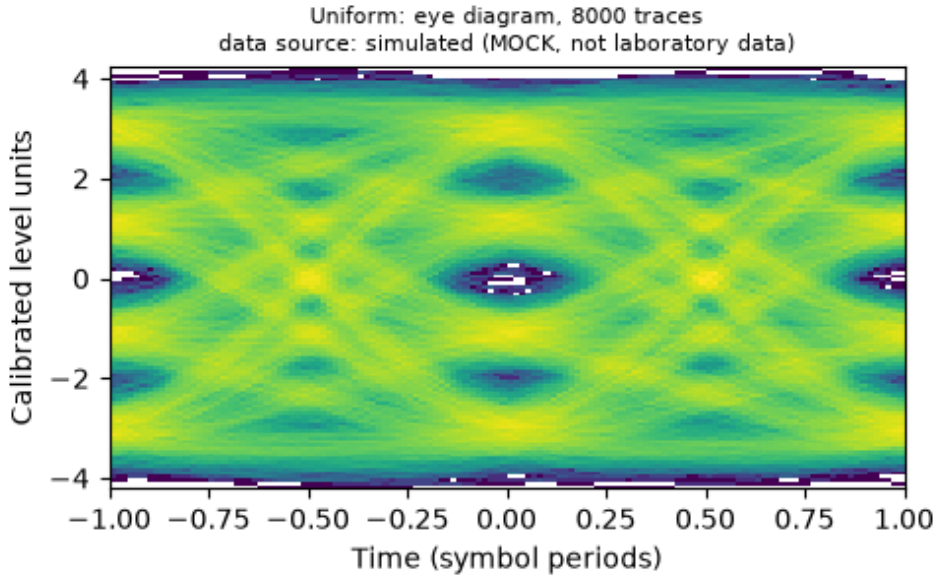
Figures



Uniform, transmitted and received waveform excerpts, capture mock-PAM4_Uniform-r1 (data source: simulated)



Uniform, recovered symbol distribution against the target, capture mock-PAM4_Uniform-r1 (data source: simulated)



Uniform, eye diagram, capture mock-PAM4_Uniform-r1 (data source: simulated)

Signal quality

Capture	SNR (dB)	R_SMD (bit/sym)	R_BMD (bit/sym)	rate_estimate_reliable
mock-PAM4_Uniform-r1	18.98	2.000	2.000	true

Capture	Level	Symbols	Mean (level units)	Spread, std (level units)	Spread, std (V, filtered-record gain)
mock-PAM4_Uniform-r1	-3	31832	-2.959	0.2468	5.923e-05
mock-PAM4_Uniform-r1	-1	32020	-1.107	0.2514	6.034e-05
mock-PAM4_Uniform-r1	+1	30888	1.096	0.2492	5.981e-05
mock-PAM4_Uniform-r1	+3	32236	2.953	0.2503	6.007e-05

Receiver acquisition configuration

Capture	Receiver model	Source format	Sample rate (GSa/s)	Bandwidth (GHz)	Matched filter (one-symbol NRZ)
mock-PAM4_Uniform-r1	MOCK-RECEIVER (no real instrument); file reports no model	csv	80 (from the file)	8 (declared)	on

Capture	Estimated symbol rate (GBd)	Clock offset (ppm)	Frames found	Frames analysed	Frames lost
mock-PAM4_Uniform-r1	8	+0.0 (confidence high)	5	4	0 (expected 4)

Simulation versus capture

Capture	Received power (dBm)	Nearest simulated power (dBm)	Power difference	Simulated SER	Capture SER
mock-PAM4_Uniform-r1	-15.41	-15.01	-0.40 dB (capture minus simulation)	6 errors in 253952 symbols, rate 2.36e-05 (95% interval 8.67e-06 to 5.14e-05)	12 errors in 126976 symbols, rate 9.45e-05 (95% interval 4.88e-05 to 0.000165); 4 frames repeat the same 31744 payload symbols

The simulated point has the same waveform and profile and the nearest received power. The mock captures use the same link configuration and profile as the simulation. The attenuation setting may differ: the simulated point is the nearest in received power, and the offset is in the table above. They also differ in the capture sample rate (the mock capture runs at the rate in the acquisition table, the simulation at 8 samples per symbol), in the receiver chain (both apply a one-symbol filter, but the capture chain reads a non-periodic record with symbol clock estimation, the simulation one exact period), and in the noise realisation.

Uncertainty

Capture	Statistical: SER 95% interval, or one-sided upper bound at 0 errors	Statistical: BER 95% interval, or one-sided upper bound at 0 errors	Statistical: frame bootstrap, SER	Physical uncertainty, from the sidecar
mock-PAM4_Uniform-r1	4.88e-05 to 0.000165	2.44e-05 to 8.25e-05	7.09e-05 to 0.000118	not recorded

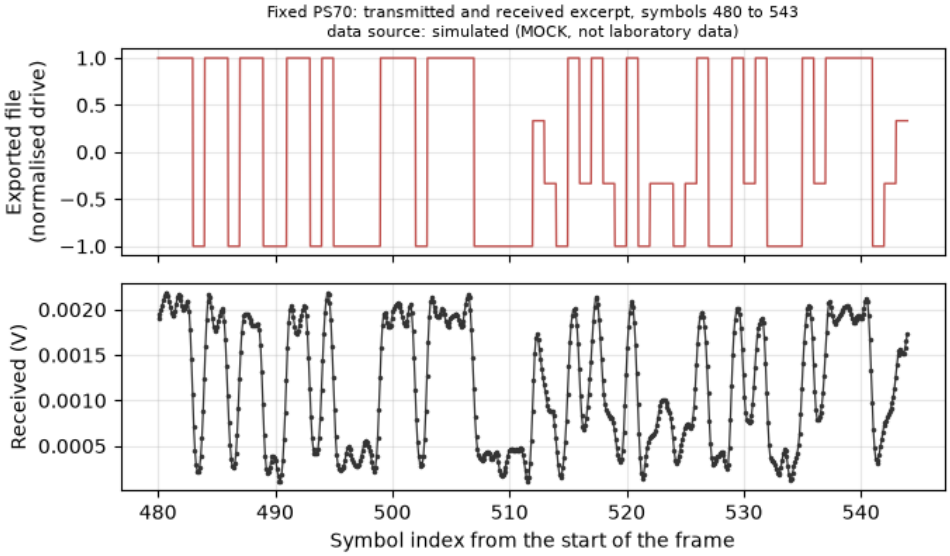
Fixed PS70 (PAM4_Fixed_PS70)

Error rates

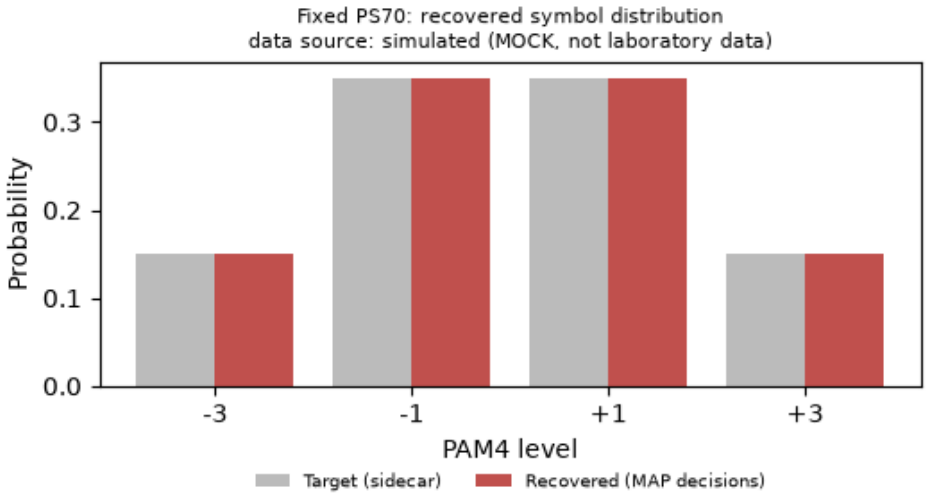
Capture	Condition	SER	BER
mock-PAM4_Fixed_PS70-r1	attenuation 18.4 dB, fiber 0 km, run r1	10 errors in 126976 symbols, rate 7.88e-05 (95% interval 3.78e-05 to 0.000145); 4 frames repeat the same 31744 payload symbols	10 errors in 253952 bits, rate 3.94e-05 (95% interval 1.89e-05 to 7.24e-05); 4 frames repeat the same 63488 payload bits

Post-matcher BER is diagnostic only; there is no FEC.

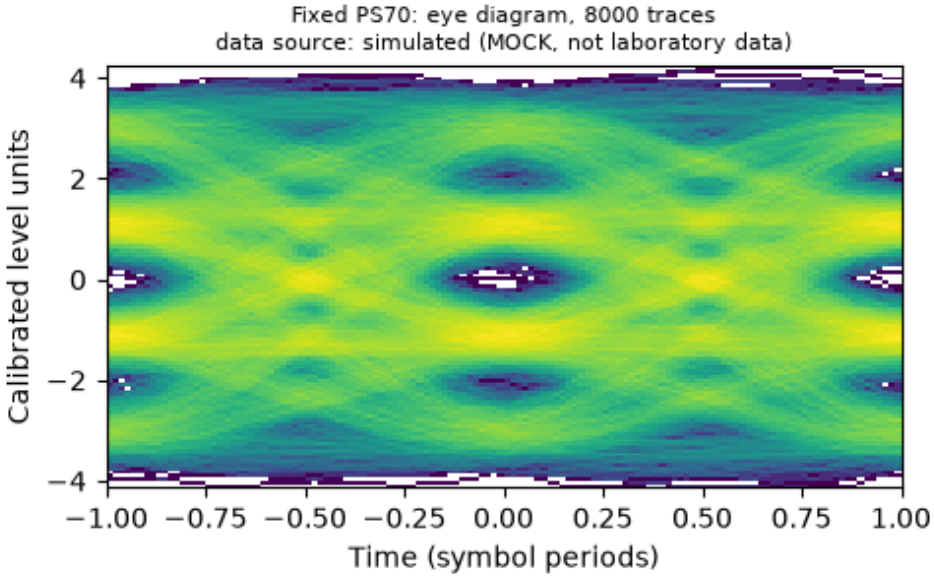
Figures



Fixed PS70, transmitted and received waveform excerpts, capture mock-PAM4_Fixed_PS70-r1 (data source: simulated)



Fixed PS70, recovered symbol distribution against the target, capture mock-PAM4_Fixed_PS70-r1 (data source: simulated)



Fixed PS70, eye diagram, capture mock-PAM4_Fixed_PS70-r1 (data source: simulated)

Signal quality

Capture	SNR (dB)	R_SMD (bit/sym)	R_BMD (bit/sym)	rate_estimate_reliable
mock-PAM4_Fixed_PS70-r1	17.43	1.881	1.881	true

Capture	Level	Symbols	Mean (level units)	Spread, std (level units)	Spread, std (V, filtered-record gain)
mock-PAM4_Fixed_PS70-r1	-3	19096	-2.963	0.2452	5.884e-05
mock-PAM4_Fixed_PS70-r1	-1	44392	-1.109	0.2479	5.949e-05
mock-PAM4_Fixed_PS70-r1	+1	44392	1.1	0.251	6.022e-05
mock-PAM4_Fixed_PS70-r1	+3	19096	2.956	0.2507	6.017e-05

Receiver acquisition configuration

Capture	Receiver model	Source format	Sample rate (GSa/s)	Bandwidth (GHz)	Matched filter (one-symbol NRZ)
mock-PAM4_Fixed_PS70-r1	MOCK-RECEIVER (no real instrument); file reports no model	csv	80 (from the file)	8 (declared)	on

Capture	Estimated symbol rate (GBd)	Clock offset (ppm)	Frames found	Frames analysed	Frames lost
mock-PAM4_Fixed_PS70-r1	8	+0.0 (confidence high)	5	4	0 (expected 4)

Simulation versus capture

Capture	Received power (dBm)	Nearest simulated power (dBm)	Power difference	Simulated SER	Capture SER
mock-PAM4_Fixed_PS70-r1	-15.41	-15.01	-0.40 dB (capture minus simulation)	3 errors in 253952 symbols, rate 1.18e-05 (95% interval 2.44e-06 to 3.45e-05)	10 errors in 126976 symbols, rate 7.88e-05 (95% interval 3.78e-05 to 0.000145); 4 frames repeat the same 31744 payload symbols

The simulated point has the same waveform and profile and the nearest received power. The mock captures use the same link configuration and profile as the simulation. The attenuation setting may differ: the simulated point is the nearest in received power, and the offset is in the table above. They also differ in the capture sample rate (the mock capture runs at the rate in the acquisition table, the simulation at 8 samples per symbol), in the receiver chain (both apply a one-symbol filter, but the capture chain reads a non-periodic record with symbol clock estimation, the simulation one exact period), and in the noise realisation.

Uncertainty

Capture	Statistical: SER 95% interval, or one-sided upper bound at 0 errors	Statistical: BER 95% interval, or one-sided upper bound at 0 errors	Statistical: frame bootstrap, SER	Physical uncertainty, from the sidecar
mock-PAM4_Fixed_PS70-r1	3.78e-05 to 0.000145	1.89e-05 to 7.24e-05	2.36e-05 to 0.000118	not recorded

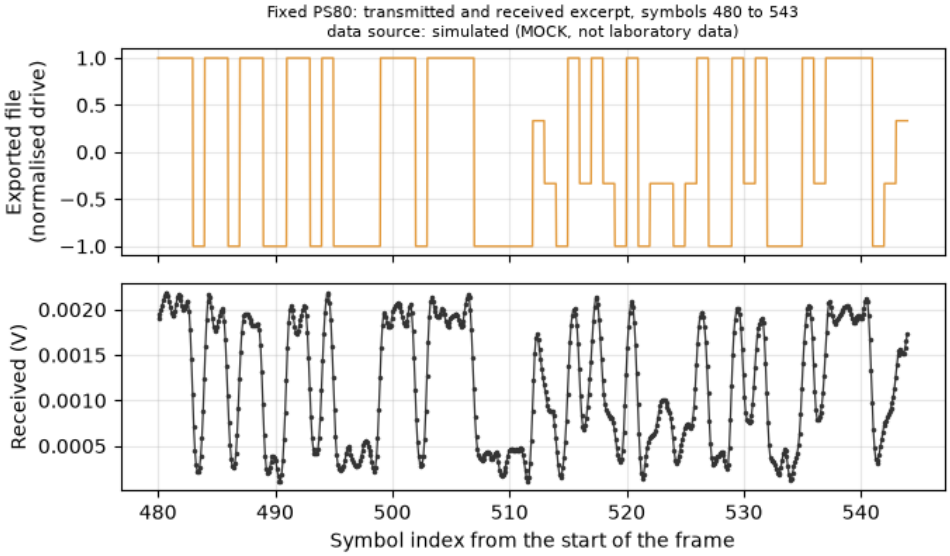
Fixed PS80 (PAM4_Fixed_PS80)

Error rates

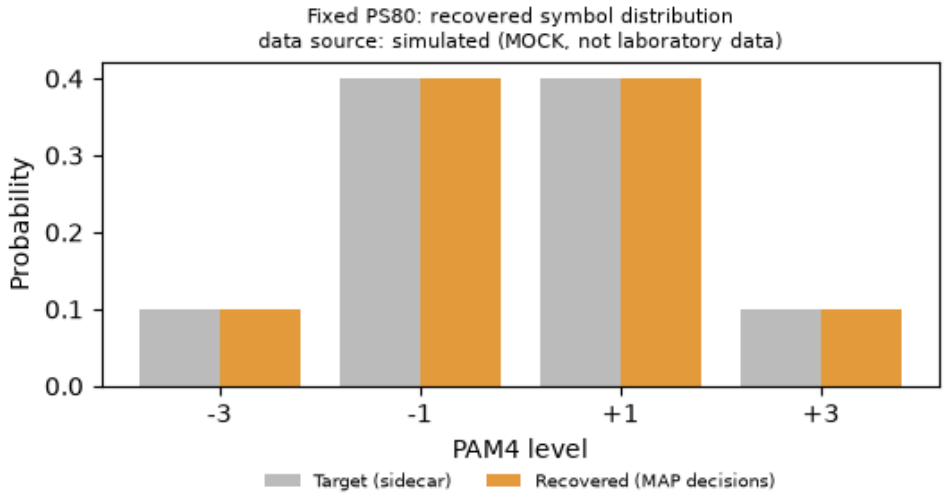
Capture	Condition	SER	BER
mock-PAM4_Fixed_PS80-r1	attenuation 18.4 dB, fiber 0 km, run r1	12 errors in 126976 symbols, rate 9.45e-05 (95% interval 4.88e-05 to 0.000165); 4 frames repeat the same 31744 payload symbols	12 errors in 253952 bits, rate 4.73e-05 (95% interval 2.44e-05 to 8.25e-05); 4 frames repeat the same 63488 payload bits

Post-matcher BER is diagnostic only; there is no FEC.

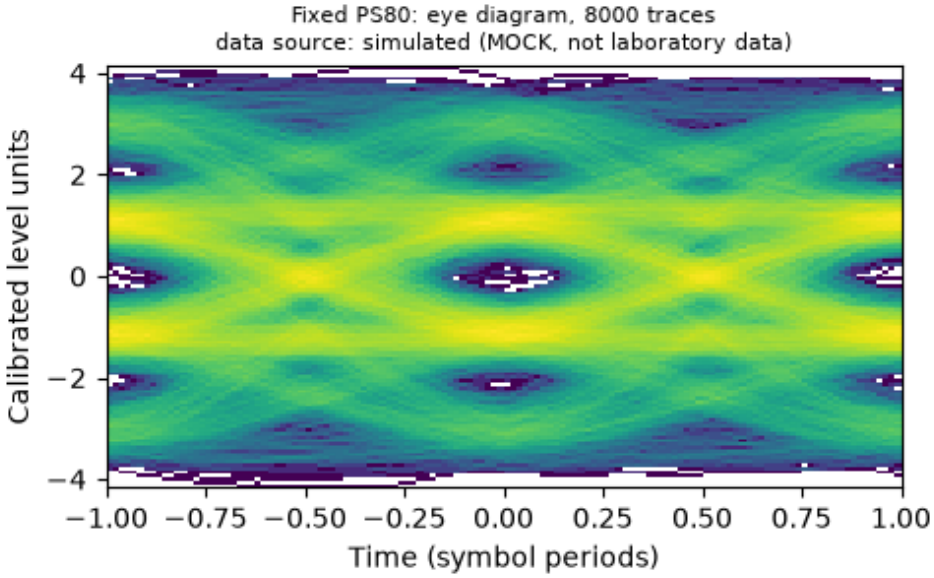
Figures



Fixed PS80, transmitted and received waveform excerpts, capture mock-PAM4_Fixed_PS80-r1 (data source: simulated)



Fixed PS80, recovered symbol distribution against the target, capture mock-PAM4_Fixed_PS80-r1 (data source: simulated)



Fixed PS80, eye diagram, capture mock-PAM4_Fixed_PS80-r1 (data source: simulated)

Signal quality

Capture	SNR (dB)	R_SMD (bit/sym)	R_BMD (bit/sym)	rate_estimate_reliable
mock-PAM4_Fixed_PS80-r1	16.38	1.722	1.722	true

Capture	Level	Symbols	Mean (level units)	Spread, std (level units)	Spread, std (V, filtered-record gain)
mock-PAM4_Fixed_PS80-r1	-3	12648	-2.967	0.2468	5.923e-05
mock-PAM4_Fixed_PS80-r1	-1	50840	-1.108	0.2475	5.94e-05
mock-PAM4_Fixed_PS80-r1	+1	50840	1.1	0.2507	6.015e-05
mock-PAM4_Fixed_PS80-r1	+3	12648	2.956	0.2488	5.97e-05

Receiver acquisition configuration

Capture	Receiver model	Source format	Sample rate (GSa/s)	Bandwidth (GHz)	Matched filter (one-symbol NRZ)
mock-PAM4_Fixed_PS80-r1	MOCK-RECEIVER (no real instrument); file reports no model	csv	80 (from the file)	8 (declared)	on

Capture	Estimated symbol rate (GBd)	Clock offset (ppm)	Frames found	Frames analysed	Frames lost
mock-PAM4_Fixed_PS80-r1	8	+0.0 (confidence high)	5	4	0 (expected 4)

Simulation versus capture

Capture	Received power (dBm)	Nearest simulated power (dBm)	Power difference	Simulated SER	Capture SER
mock-PAM4_Fixed_PS80-r1	-15.41	-15.01	-0.40 dB (capture minus simulation)	2 errors in 253952 symbols, rate 7.88e-06 (95% interval 9.54e-07 to 2.84e-05)	12 errors in 126976 symbols, rate 9.45e-05 (95% interval 4.88e-05 to 0.000165); 4 frames repeat the same 31744 payload symbols

The simulated point has the same waveform and profile and the nearest received power. The mock captures use the same link configuration and profile as the simulation. The attenuation setting may differ: the simulated point is the nearest in received power, and the offset is in the table above. They also differ in the capture sample rate (the mock capture runs at the rate in the acquisition table, the simulation at 8 samples per symbol), in the receiver chain (both apply a one-symbol filter, but the capture chain reads a non-periodic record with symbol clock estimation, the simulation one exact period), and in the noise realisation.

Uncertainty

Capture	Statistical: SER 95% interval, or one-sided upper bound at 0 errors	Statistical: BER 95% interval, or one-sided upper bound at 0 errors	Statistical: frame bootstrap, SER	Physical uncertainty, from the sidecar
mock-PAM4_Fixed_PS80-r1	4.88e-05 to 0.000165	2.44e-05 to 8.25e-05	7.09e-05 to 0.000118	not recorded

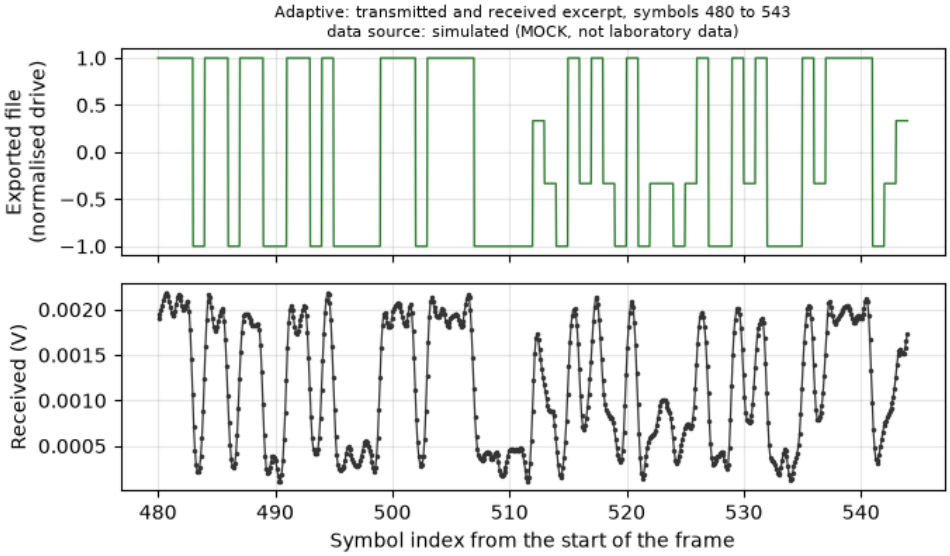
Adaptive (PAM4_Adaptive)

Error rates

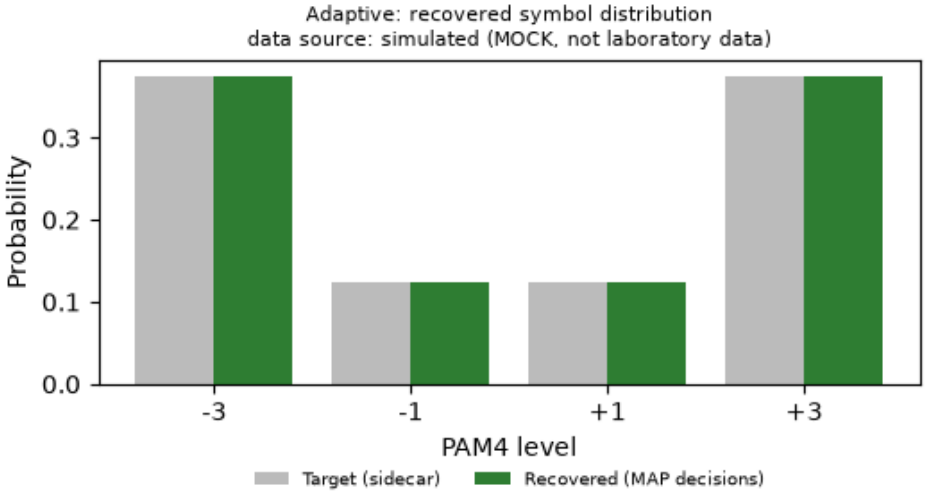
Capture	Condition	SER	BER
mock-PAM4_Adaptive-r1	attenuation 18.4 dB, fiber 0 km, run r1	11 errors in 126976 symbols, rate 8.66e-05 (95% interval 4.32e-05 to 0.000155); 4 frames repeat the same 31744 payload symbols	11 errors in 253952 bits, rate 4.33e-05 (95% interval 2.16e-05 to 7.75e-05); 4 frames repeat the same 63488 payload bits

Post-matcher BER is diagnostic only; there is no FEC.

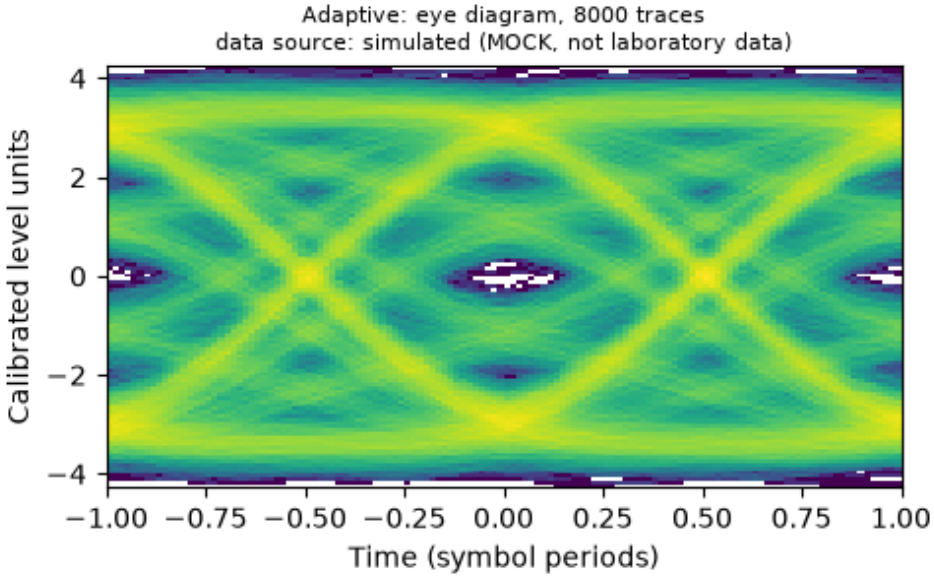
Figures



Adaptive, transmitted and received waveform excerpts, capture mock-PAM4_Adaptive-r1 (data source: simulated)



Adaptive, recovered symbol distribution against the target, capture mock-PAM4_Adaptive-r1 (data source: simulated)



Adaptive, eye diagram, capture mock-PAM4_Adaptive-r1 (data source: simulated)

Signal quality

Capture	SNR (dB)	R_SMD (bit/sym)	R_BMD (bit/sym)	rate_estimate_reliable
mock-PAM4_Adaptive-r1	20.36	1.811	1.811	true

Capture	Level	Symbols	Mean (level units)	Spread, std (level units)	Spread, std (V, filtered-record gain)
mock-PAM4_Adaptive-r1	-3	47616	-2.959	0.2495	5.988e-05
mock-PAM4_Adaptive-r1	-1	15872	-1.107	0.2461	5.904e-05
mock-PAM4_Adaptive-r1	+1	15872	1.096	0.2527	6.063e-05
mock-PAM4_Adaptive-r1	+3	47616	2.951	0.2504	6.009e-05

Receiver acquisition configuration

Capture	Receiver model	Source format	Sample rate (GSa/s)	Bandwidth (GHz)	Matched filter (one-symbol NRZ)
mock-PAM4_Adaptive-r1	MOCK-RECEIVER (no real instrument); file reports no model	csv	80 (from the file)	8 (declared)	on

Capture	Estimated symbol rate (GBd)	Clock offset (ppm)	Frames found	Frames analysed	Frames lost
mock-PAM4_Adaptive-r1	8	+0.0 (confidence high)	5	4	0 (expected 4)

Simulation versus capture

Capture	Received power (dBm)	Nearest simulated power (dBm)	Power difference	Simulated SER	Capture SER
mock-PAM4_Adaptive-r1	-15.41	-15.01	-0.40 dB (capture minus simulation)	4 errors in 253952 symbols, rate 1.58e-05 (95% interval 4.29e-06 to 4.03e-05)	11 errors in 126976 symbols, rate 8.66e-05 (95% interval 4.32e-05 to 0.000155); 4 frames repeat the same 31744 payload symbols

The simulated point has the same waveform and profile and the nearest received power. The mock captures use the same link configuration and profile as the simulation. The attenuation setting may differ: the simulated point is the nearest in received power, and the offset is in the table above. They also differ in the capture sample rate (the mock capture runs at the rate in the acquisition table, the simulation at 8 samples per symbol), in the receiver chain (both apply a one-symbol filter, but the capture chain reads a non-periodic record with symbol clock estimation, the simulation one exact period), and in the noise realisation.

Uncertainty

Capture	Statistical: SER 95% interval, or one-sided upper bound at 0 errors	Statistical: BER 95% interval, or one-sided upper bound at 0 errors	Statistical: frame bootstrap, SER	Physical uncertainty, from the sidecar
mock-PAM4_Adaptive-r1	4.32e-05 to 0.000155	2.16e-05 to 7.75e-05	6.3e-05 to 0.00011	not recorded

Comparison verdicts

- No pair received a verdict.

Withheld or absent verdicts, and reasons

- A_initial_8GBd, attenuation 18.4 dB, fiber 0 km: PAM4_Adaptive vs PAM4_Fixed_PS70: insufficient paired runs (need 5) (paired runs: 1).
- A_initial_8GBd, attenuation 18.4 dB, fiber 0 km: PAM4_Adaptive vs PAM4_Fixed_PS80: insufficient paired runs (need 5) (paired runs: 1).
- A_initial_8GBd, attenuation 18.4 dB, fiber 0 km: PAM4_Adaptive vs PAM4_Uniform: insufficient paired runs (need 5) (paired runs: 1).
- A_initial_8GBd, attenuation 18.4 dB, fiber 0 km: PAM4_Fixed_PS70 vs PAM4_Fixed_PS80: insufficient paired runs (need 5) (paired runs: 1).
- A_initial_8GBd, attenuation 18.4 dB, fiber 0 km: PAM4_Fixed_PS70 vs PAM4_Uniform: insufficient paired runs (need 5) (paired runs: 1).
- A_initial_8GBd, attenuation 18.4 dB, fiber 0 km: PAM4_Fixed_PS80 vs PAM4_Uniform: insufficient paired runs (need 5) (paired runs: 1).

Flagged and excluded captures

- No capture is flagged reference_mismatch_suspected.
- Excluded from the rate comparison because the rate estimate is unreliable: none.

Set-point audit

Variable	Role	Status	Range
Symbol rate	set-point	controlled	8e+09 to 8e+09
Capture sample rate	set-point	controlled	8e+10 to 8e+10
AWG sample rate	set-point	controlled	2.24e+11 to 2.24e+11
Electrical amplitude	set-point	controlled	2 to 2
Optical launch power	outcome	reported	2.98939 to 2.98978
Optical received power	outcome	controlled	-15.4106 to -15.4102
Optical modulation amplitude	outcome	reported	4.72564 to 4.72696
Extinction ratio	outcome	reported	8.37097 to 8.37209
Fiber length	set-point	controlled	0 to 0
Attenuation	set-point	controlled	18.4 to 18.4
Receiver bandwidth	set-point	controlled	8e+09 to 8e+09

Set-points must match for a verdict. Outcomes differ between waveforms at a fixed drive by design and are reported, never gating, except received power beyond its tolerance (0.5 dB).

Uncertainty

Statistical uncertainty is the sampling variation of the error counts: Clopper-Pearson 95% intervals over the pooled symbols, which assume independent errors, and a bootstrap over frames. Frames repeat one payload, so these intervals are optimistic and describe noise only. Physical measurement uncertainty (instrument calibration, power meter, attenuator and bandwidth accuracy) is a separate matter, is not part of any interval above, and is not recorded unless the sidecar's calibration field gives it. For these captures it is stated per capture in the uncertainty tables.

Limitations

- No real UXR export has been imported by this project; the importer follows the vendor guide and was tested on synthetic files. Treat every capture here accordingly.
- Sample loss of less than about 1 symbol, synchronous with the frame period on one waveform in every run, may not be detected.
- Lower error rate or lower symbol energy is not a throughput gain.